Communicating with a Drone using Python and MQTT

Multi-ESP32 MQTT Drone Control System

Overview

This project establishes a multi-layer MQTT communication system connecting a Python script, two ESP32 boards, and a Pluto Drone. It demonstrates real-time control of drone operations such as **arm**, **takeoff**, **land**, and **disarm** over Wi-Fi using MQTT topics. The setup helps in understanding distributed communication systems for IoT and drone-based automation.

1) Hardware Requirements

Component	Quantity	Description
ESP32	2	Wi-Fi enabled microcontrollers used as bridge and listener.
Pluto Drone	1	Target drone to receive and execute commands.

System Topology

Python Script → ESP32 (Bridge / Hub) → ESP32 (Listener) → Pluto Drone
(MQTT) (Wi-Fi) (Serial / Command Execution)
All communication between devices is handled using the MQTT protocol.

Component Description

1. ESP32 – MQTT Bridge (Wi-Fi Hub)
• Acts as a Wi-Fi Access Point (AP).
• Subscribes to topic: **pluto/cmd/#**
• Forwards all received messages to: **bridge/esp8266/cmd**
• Displays logs and messages via the Serial Monitor.
2. ESP32 – MQTT Listener
• Connects to the ESP32-Hub Wi-Fi network.
• Subscribes to topic: **bridge/esp32/cmd**
• Receives messages and executes corresponding drone commands:
a) arm
b) disarm
c) takeoff
d) land
3. Python – Command Publisher
• Script File: **send_mqtt_cmd.py**
• Publishes MQTT messages to the drone topics:
pluto/cmd/arm
pluto/cmd/disarm
pluto/cmd/takeoff
pluto/cmd/land

Run Command

To execute drone commands via Python, run the following command:

```
python3 send_mqtt_cmd.py arm
```

Replace “arm” with desired command: **disarm**, **takeoff**, or **land**.

Command Execution via Terminal

Once all connections are established, you can manually send MQTT commands using **mosquitto_pub**:

```
mosquitto_pub -h 192.168.4.2 -t "pluto/cmd/arm" -m "arm"
mosquitto_pub -h 192.168.4.2 -t "pluto/cmd/takeoff" -m
"takeoff"
mosquitto_pub -h 192.168.4.2 -t "pluto/cmd/land" -m "land"
```

Troubleshooting & Network Utilities

Checking Local IP Address

To identify your local IP (Mac/Linux), use:

```
ipconfig getifaddr en0
```

Common Issues

Issue	Possible Cause	Solution
MQTT messages not received	Incorrect topic or broker IP	Verify MQTT topics and broker IP
ESP32 not connecting	Wrong SSID/password	Double-check Wi-Fi credentials
No serial output	Baud rate mismatch	Ensure Serial Monitor is set to correct baud ra

Future Improvements

- Add telemetry feedback from the drone to the Python dashboard.
- Include command acknowledgment for improved reliability.
- Develop a web-based MQTT interface for easier user interaction.
- Integrate visual feedback via onboard camera stream for live operation.

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